

Comparative Evaluation of Neural Architectures for Sentiment Analysis on Financial News: From MLPs to Transformers

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1 Introduction

Traditional financial models are fully reliant on quantitative inputs such as stock prices and macroeconomic indicators. Yet, these indicators rarely capture the full picture of market behaviour. In recent years, there has been growing interest in integrating sentiment analysis to account for the emotional and psychological factors that influence investor decisions (Atkins, 2018). Among text sources social media has received the majority of research within the industry, on the other hand, news content has received less attention despite its proven potential (Ashtiani, 2023).

While traditional machine learning (ML) methods provided a foundation for sentiment analysis, they underperform in more complex domains like finance. ML models rely on domain-specific feature engineering limiting their ability to understand complex or subtle sentiments which are frequent in financial news (Bashiri and Naderi, 2024). This can also introduce bias, reducing generalisability and their power in unseen datasets. Deep learning methods such as RNNs, LSTMs and Transformers offer a more accurate alternative, stemming from their ability to capture sequential patterns, making them better suited for analysing sentiment in financial news (Bashiri and Naderi, 2024).

FinBERT, developed by Huang, Wang, and Yang (2022), is a finance-specific language model based on BERT but trained on financial documents and news. It significantly outperforms general-purpose models and other ML methods, e.g. BERT, Naive Bayes and LSTMs, in extracting sentiment and insights from financial text, proving that these more complex methods hold value in sentiment analysis and stock market prediction.

This project compares multiple neural architectures to evaluate which best captures financial sentiment from news headlines, with the aim of identifying patterns that could identify market movement more effectively than traditional data alone.

2 Dataset Description

This project uses the *English Financial News* dataset (Calarte, 2023), which contains approximately 27,000 financial news headlines collected from Twitter and Yahoo Finance. Each headline is labelled as *positive*, *negative*, or *neutral* according to its expressed sentiment.

The dataset was selected due to its large and diverse corpus of real-world financial language, which closely represents the types of news and social media content that influence investor sentiment and market movements.

Given the unstructured nature of textual data, preprocessing was required before training the models. The following steps were applied:

- Removal of URLs, stock tickers, special symbols, and extraneous whitespace.
- Tokenisation and vectorisation of each sentence to convert text into numerical form suitable for neural network input.

The dataset was split into training (80%) and testing (20%) subsets to enable model evaluation. While well-suited for sentiment classification tasks, the dataset exhibits a mild class imbalance, with *neutral* sentiment being most common. This imbalance was accounted for during model training through appropriate sampling and weighting techniques.

3 Methods

This project defines financial sentiment prediction as a three-class classification problem, labelling each headline as positive, neutral or negative. A baseline TF-IDF + Logistic Regression model was

compared with three neural architectures (MLP, LSTM and Transformer) trained on FinBERT contextual embeddings from ProsusAI (2023).

The baseline used TF-IDF features and classified using Logistic Regression, which is commonly included in previous research (Karanikola et al., 2023)

Each headline was encoded using FinBERT (Huang, Wang and Yang, 2022), which generates contextual and positional embeddings, as it had been shown to outperform general-purpose models in financial sentiment tasks (Huang, Wang and Yang, 2022). These embeddings were then passed through three architectures: an MLP with ReLU activation (Goodfellow et al., 2023), a bidirectional LSTM to model sequential relationships (Sharaff et al., 2023), and a lightweight Transformer with self-attention (Vaswani et al., 2017).

All models were trained on 80% of the data with early stopping and evaluated on the remaining 20% using various performance metrics. Full details on training parameters, metrics and architecture configurations are provided in the appendix.

4 Results

Summarise main findings, supported by figures and tables. Place figures in the third page of the report using `\begin{figure}...\end{figure}` and reference them in text using `\ref{}`. For instance, see Figure ??.

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5 Discussions

Discuss results interpretation, advantages/disadvantages of choices, trade-offs, and limitations.

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6 Conclusions

Summarise key takeaways and possible future work.

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Link to Presentation

<http://url.to.your.video.presentation/make-sure-permissions-are-granted>

Appendix A - Extended Methods

Table 1: Summary of Methods and Model Architectures

Component	Description / Configuration
Preprocessing	• Lowercased all text • Removed URLs, stock tickers (e.g., \$AAPL), special characters • Removed extra whitespace
Tokenisation and Embeddings	• Used <code>ProsusAI/finbert</code> (ProsusAI, 2023) tokenizer from Hugging Face • Truncated input sequences to a maximum length of 64 tokens • No padding at tokenisation step; applied dynamically during batching • Extracted 768-dimensional [CLS] embeddings from FinBERT
Model Architectures	• Baseline: TF-IDF + Logistic Regression • MLP: Three fully connected layers ($768 \rightarrow 256 \rightarrow 128 \rightarrow 64$) with ReLU activations, dropout (0.3), and LayerNorm; output fed to a softmax classifier. • LSTM: Three-layer bidirectional LSTM (hidden size = 256) with dropout (0.3), followed by LayerNorm, an attention layer, and a final classification layer for 3-class output. • Transformer: Four-layer Transformer Encoder with 8 attention heads, feedforward dim = 256, LayerNorm before/after encoding, followed by a two-layer MLP classifier. • All models use FinBERT contextual embeddings as frozen input features.
Training Process	• Loss Function - CrossEntropy Loss with L2 Regularisation and Label Smoothing • Optimiser - AdamW with learning rate of 1×10^{-4} • Gradient clipping at max norm of 1.0 to prevent exploding gradients • Applied early stopping with a patience of 5 epochs (monitored validation loss) • Trained for a maximum of 100 epochs with batch size of 64 • FinBERT parameters frozen to reduce training time and memory usage